SHALOM HUNUKUMBURA

FULL STACK DEVELOPER • PYTHON · FASTAPI · NEXT.JS · TYPESCRIPT · AI/LLM

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PROFESSIONAL SUMMARY

- **Full Stack Developer** specializing in Python, FastAPI, React/Next.js and TypeScript, with professional experience building production grade business, taxation and compliance platforms.
- Delivers features **end to end**: REST API design, relational data modeling, frontend implementation, deployment and production incident response.
- Builds **AI powered applications** using Retrieval-Augmented Generation (RAG), LLM integration and prompt engineering, improving contextual accuracy for finance and compliance use cases.
- Experienced in **production troubleshooting, CI/CD and database migrations**, resolving live incidents with minimal downtime.
- Ships **open-source tools in public**: 4 Python packages on PyPI, an MCP server on the official MCP Registry, and Unlisted, a live job search product built on 7,700+ company career pages.

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript, SQL, HTML, CSS
Backend	FastAPI, Flask, SQLAlchemy, Node.js, Express, REST API design
Frontend	React, Next.js, TypeScript, Tailwind CSS, responsive UI development
Databases	PostgreSQL, MySQL, MongoDB, SQLite, schema design & migrations
AI & ML	Retrieval-Augmented Generation (RAG), LLM integration, prompt engineering, embeddings & vector retrieval (FAISS), hybrid search (BM25 + RRF), RAG evaluation, MCP servers (Model Context Protocol), agentic AI workflows, Scikit-learn
Cloud & DevOps	AWS (S3, core infrastructure concepts), Linux (Ubuntu), GitHub Actions CI/CD, Vercel, Neon Postgres, Docker, PyPI packaging & trusted publishing, deployment debugging
Tools	Git, Postman, VS Code, JWT authentication, AI-assisted development tooling

PROFESSIONAL EXPERIENCE

Software Engineer, Full-Stack Developer

Aug 2025 – Present

Xaventra | BizAdvisor Platform

- Build and ship features across a 15-service production platform covering taxation, accounting, incorporation and compliance workflows, authoring 170+ merged pull requests on a 6-engineer team.
- Design and integrate RESTful APIs across FastAPI and Flask services backed by PostgreSQL, contributing to 140+ endpoints consumed by Next.js and TypeScript frontends.
- Own features end to end across 12 services, from database schema and backend business logic through frontend implementation to production release, delivering invoicing, payments, promotions and incorporation workflows.
- Diagnose and resolve production incidents through log analysis, database inspection and API tracing, restoring service with minimal downtime and documenting root causes.
- Author 46 Alembic database migrations and maintain 8 GitHub Actions CI/CD workflows, including automated migration checks, ensuring safe repeatable deployments.
- Collaborate with backend and frontend engineers across sprint cycles, contributing 380+ commits and reviewing changes across the platform's repositories.

Software Engineering Intern, Full-Stack Developer **Feb 2025 – Aug 2025**
Xaventra | BizAdvisor & Neurux Platforms

BizAdvisor | *compliance platform for corporate secretarial firms*

- Developed scalable backend services with Flask and SQLAlchemy, integrated with React and TypeScript frontends to support complex multi-party task coordination workflows.
- Diagnosed and fixed intricate ORM relationship and data-binding defects, correcting coordinator attribution and eliminating inaccurate UI rendering.
- Refactored legacy components and resolved production issues in an agile environment, improving maintainability and system reliability.

Neurux | *domain-focused AI assistant for business & compliance*

- Adapted Neurux, an AnythingLLM fork, into a domain-specific AI chatbot for finance and compliance.
- Implemented Retrieval-Augmented Generation (RAG) pipelines with embeddings and structured document retrieval, measurably improving contextual relevance and response precision.
- Iterated on prompt structures to reduce off-domain and inaccurate responses, and prototyped agentic workflows for multi-step tasks.

SELECTED PROJECTS

Unlisted | Job Search Across 7,700+ Company Career Pages [Live] [GitHub]
Next.js 16, TypeScript, Python, PostgreSQL (Neon), GitHub Actions, Vercel

- Live search over every role on 7,700+ Greenhouse, Ashby and Lever career pages, synced hourly, keeping only the past two weeks (~40,000 of 227,000 jobs).
- Classifies where a remote role can really be done from (anywhere, a region or one country) and infers experience level.
- Cut storage from 857 MB to under 400 MB to fit a 512 MB free Postgres, with searches in milliseconds; cut the home page from 659 KB to 133 KB with caching.

lk-tax & lk-tax-mcp | Sri Lankan Tax Engine and AI Tool Server [Live] [GitHub] [PyPI] [MCP]
Python, Decimal, FastAPI, Pyodide (WebAssembly), MCP Python SDK

- Engine for APIT, income tax, WHT, VAT and SSCL with rules stored as dated data citing Inland Revenue sources, so any date since 2023 uses the rules then in force.
- Verified against all 75,590 rows of the IRD's official salary tax tables (134 tests); runs fully in the browser as a live calculator via Pyodide.
- MCP server on the official MCP Registry giving AI assistants 9 exact, sourced tax tools instead of recalled rates.

alembic-guard | Migration Safety Linter for PostgreSQL [GitHub] [PyPI]
Python, AST, Alembic, PostgreSQL, GitHub Actions

- CLI and GitHub Action (PyPI, GitHub Marketplace) that catches Alembic migrations likely to lock Postgres tables or break running code, using Python's `ast`.
- 10 rules with concrete zero-downtime fixes, diff-only pull request checks and inline GitHub annotations; a demo repository shows an unsafe migration blocked in a PR.

payhere-sim | Local Testing Tool for PayHere Payments [GitHub] [PyPI]
Python (standard library only), webhooks, HTTP

- Signs payment notifications exactly as Sri Lanka's PayHere gateway does and delivers them to localhost.
- Checks a webhook handler against 10 scenarios (forged, underpaid, duplicate, late and simultaneous notifications); caught a race that granted 500 credits for one payment in a handler that passed manual testing.

DocSage | Retrieval-Augmented Question Answering from First Principles [GitHub]
Python, FAISS, BM25, sentence-transformers, FastAPI, Docker

- RAG without LangChain over 37 documents with citations; hybrid FAISS + BM25 retrieval lifted top-1 accuracy to 93% from 86%.

EDUCATION & CERTIFICATIONS

BSc (Hons) in Information Technology, Cloud Computing · *Sri Lanka Technological Campus* **2025**
AWS Cloud Practitioner Training **Completed**